

Single line sftp from terminal

Asked 12 years, 5 months ago Modified 6 months ago Viewed 236k times



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Several times throughout the day, I may be running a test where I need to look through a log file on a remote server. I've gotten used to using my terminal to `sftp` into the remote server and pull the desired log file down to `/tmp` on my local machine.



I was looking through the options today using `man sftp` in an attempt to figure out a way to run the following commands basically in a single line so that I don't have to type a command, press enter, type a command press enter, etc.



(what I do now)

```
sftp myuser@myserver
--mypassword at prompt
lcd /tmp
get /dir/dir/dir/dir/file
quit
```

I found while looking through `man sftp` a reference to `scp` which I haven't used before. I feel it may be what I'm looking for, but I didn't see a way to specify where I wanted the securely copied file to go.

Could someone provide me with a way to get `/dir/file` from a remote server and have it download to `/tmp/file_plus-my-description`?

I was hoping to be able to run an `sftp` or `scp` command similar to a regular UNIX copy like:

```
scp myuser@myserver /dir/file /tmp/file_plus-my-description
```

I'm using the built in `Terminal` in Mac OS X 10.8. Thanks.

unix

terminal

sftp

scp

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edited Jan 9, 2021 at 7:28



Martin Prikryl

205k 64 559 1.1k

asked May 23, 2013 at 19:02



Kyle

14.8k 12 49 62



A related question on Super User: [Using sftp like scp.](#) – Martin Prikryl Jan 22, 2021 at 13:59

8 Answers

Sorted by:

Highest score (default)



Update Sep 2017 - tl;dr



Download a single file from a remote ftp server to your machine:

286

```
sftp {user}@{host}:{remoteFileName} {localFileName}
```



Upload a single file from your machine to a remote ftp server:



```
sftp {user}@{host}:{remote_dir} <<< '$put {local_file_path}'
```



Original answer:

Ok, so I feel a little dumb. But I figured it out. I almost had it at the top with:

```
sftp user@host remoteFile localFile
```

The only documentation shown in the terminal is this:

```
sftp [user@]host[:file ...]
sftp [user@]host[:dir[]]
```

However, I came across [this site](#) which shows the following under the synopsis:

```
sftp [-vC1 ] [-b batchfile ] [-o ssh_option ] [-s subsystem | sftp_server ] [-B
buffer_size ] [-F ssh_config ] [-P sftp_server path ] [-R num_requests ] [-S
program ] host
sftp [[user@]host[:file [file]]]
sftp [[user@]host[:dir[]]]
```

So the simple answer is you just do : after your user and host then the remote file and local filename. Incredibly simple!

Single line, sftp copy remote file:

```
sftp username@hostname:remoteFileName localFileName
sftp kyle@kylesserver:/tmp/myLogFile.log /tmp/fileNameToUseLocally.log
```

Update Feb 2016

In case anyone is looking for the command to do the reverse of this and *push* a file from your local computer to a remote server in one single line `sftp` command, user @Thariama below posted the solution to accomplish that. Hat tip to them for the extra code.

```
sftp {user}@{host}:{remote_dir} <<< '$put {local_file_path}'
```

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edited Oct 25, 2018 at 19:51

answered May 23, 2013 at 20:19



Kyle

14.8k

12

49

62

! You can now comment to request clarification or add additional context.



Comments



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16



OpenSSH `scp` [supports SFTP protocol since 8.7](#).

Since [OpenSSH 9.0](#) the `scp` uses SFTP by default.

In 8.7 through 8.9, the SFTP has to be selected via `-s` parameter.

Download:



```
scp -s user@host:/remote/source/path /local/target/path
```

Upload:

```
scp -s /local/source/path user@host:/remote/target/path
```

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edited May 23, 2022 at 6:18

answered Aug 8, 2021 at 13:15



Martin Prikryl

205k

64

559

1.1k

Comments



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16



SCP answer

The OP mentioned SCP, so here's that.

As others have pointed out, SFTP is a confusing since the *upload* syntax is completely different from the *download* syntax. It gets *marginally* easier to remember if you use the same form:



```
echo 'put LOCALPATH REMOTEPATH' | sftp USER@HOST
echo 'get REMOTEPATH LOCALPATH' | sftp USER@HOST
```

In reality, this is still a mess, and is why people still use "outdated" commands such as SCP:

```
scp USER@HOST:REMOTEPath LOCALPath
scp LOCALPath USER@HOST:REMOTEPath
```

SCP is secure but dated. It has some bugs that will never be fixed, namely crashing if the server's `.bash_profile` emits a message. However, in terms of usability, the devs were years ahead.

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edited Nov 4, 2020 at 11:14

answered Nov 4, 2020 at 11:04



c z

9,308

4

58

63

Comments



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8

Or echo 'put {path to file}' | sftp {user}@{host}:{dir} , which would work in both unix and powershell.

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answered Nov 9, 2019 at 0:26



js2010

28.2k

6

82

86



1 Comment ▾



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9 Guy Over a year ago

This syntax also works with puttyftp (psftp). Just thought I'd share since that's the reason I ended up here

▲ 0

Reply



7

To UPLOAD a single file, you will need to create a bash script. Something like the following should work on OS X if you have `sshpass` installed.

Usage:

```
sftpx <password> <user@hostname> <localfile> <remotefile>
```



Put this script somewhere in your path and call it `sftpx` :

```
#!/bin/bash
```

```
export RND=`cat /dev/urandom | env LC_CTYPE=C tr -cd 'a-f0-9' | head -c 32`
export TMPDIR=/tmp/$RND
```

```
export FILENAME=$(basename "$4")
export DSTDIR=$(dirname "$4")

mkdir $TMPDIR
cp "$3" $TMPDIR/$FILENAME

export SSHPASS=$1
sshpass -e sftp -oBatchMode=no -b - $2 << !
    lcd $TMPDIR
    cd $DSTDIR
    put $FILENAME
    bye
!

rm $TMPDIR/$FILENAME
rmdir $TMPDIR
```

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answered Dec 23, 2014 at 4:55

[sstur](#)

1,819 21 22

2 Comments ▾



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[Kyle](#) Over a year ago

Thanks for the reply, however, the answer I denoted above will utilize single line command with sftp to get a file from a remote server and save the file to a specific location on your local machine. I have verified that it will work on OS X with only sftp.

▲ 0

Reply

[sstur](#) Over a year ago

Yes, that is certainly the best answer for transferring from remote to local. However I though I'd add a solution for doing the opposite: local to remote. Just in case it's helpful to anyone. Thanks!

▲ 2

Reply



sftp supports batch files.

4

From the man page:



-b batchfile



Batch mode reads a series of commands from an input batchfile instead of stdin. Since it lacks user interaction it should be used in conjunction with non-interactive authentication. A batchfile of '-' may be used to indicate standard input. sftp will abort if any of the following commands fail: get, put, rename, ln, rm, mkdir, chdir, ls, lcdir, chmod, chown, chgrp, lpwd, df, symlink, and lmkdir.

Termination
on error can be suppressed on a command by command basis by prefixing the command with a '-' character (for example, -rm /tmp/blah*).

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answered May 23, 2013 at 19:08



synthesizerpatel

28.3k 5 76 92

2 Comments ▾



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Kyle Over a year ago

As this is a process I do several times a day for different files in different places, it would be nice to be able to just memorize a command I could type into a single line in my terminal (which is always open) and press enter. Using `--batch` would require me to put my commands into a file then pass in that file as a param. I'm still playing with `scp`, I feel it may contain my solution.

▲ 0

Reply



synthesizerpatel Over a year ago

Maybe 'expect' is what you're looking for then. linuxjournal.com/article/3065

▲ 0

Reply



2



A minor modification like below worked for me when using it from within perl and system() call:

```
sftp {user}@{host} <<< '$put {local_file_path} {remote_file_path}'
```

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answered Jun 9, 2020 at 12:36



user2443447

149 1 8

Comments



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0



Not the most secure solution, but if you need a one-liner including the password...

```
sshpass -p <password> sftp <user>@<destination> <<< '$put <filepath>'
```

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answered Apr 17 at 19:39



Evelyn Dickinson

21 1 4

Comments



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